

CENG 3420 Lab2-1 Report

Name: Wong Kwok Shing SID:1155234495

Step by Step of making this simulator

1. Making the decoder

```
/*
 * Lab2-2 assignment: Decode other types of RV32I instructions
 */
case 0x37:
    handle_lui(cur_inst);
    break;

case (0x0C << 2) + 0x03:
    switch (funct3)
    {
        case 0:
            if (MASK31_25(cur_inst) == 0)
                handle_add(cur_inst);
            else
                handle_sub(cur_inst);
            break;
        case 1:
            handle_sll(cur_inst);
            break;
        case 4:
            handle_xor(cur_inst);
            break;
        case 5:
            if (MASK31_25(cur_inst) == 0)
                handle_srl(cur_inst);
            else
                handle_sra(cur_inst);
            break;
        case 6:
            handle_or(cur_inst);
            break;
        case 7:
            handle_and(cur_inst);
            break;
        default:
            error("unknown opcode 0x%08x is captured.\n", cur_inst);
    }
    break;

case 0x67:
    handle_jalr(cur_inst);
    break;
case 0x6F:
    handle_jal(cur_inst);
    break;
case (0x18 << 2) + 0x03:
    switch (funct3)
    {
        case 0x0:
            handle_beq(cur_inst);
            break;
        case 0x1:
            handle_bne(cur_inst);
            break;
        case 0x4:
            handle_blt(cur_inst);
            break;
        case 0x5:
            handle_bge(cur_inst);
            break;
        default:
            error("unknown opcode 0x%08x is captured.\n", cur_inst);
    }
    break;
case 0x03:
    switch (funct3)
    {
        case 0x0:
            handle_lb(cur_inst);
            break;
        case 0x1:
            handle_lh(cur_inst);
            break;
        case 0x2:
            handle_lw(cur_inst);
            break;
    }
    break;

case 0x23:
    switch (funct3)
    {
        case 0x0:
            handle_sb(cur_inst);
            break;
        case 0x1:
            handle_sh(cur_inst);
            break;
        case 0x2:
            handle_sw(cur_inst);
            break;
    }
    break;
default:
    error("unknown instruction 0x%08x is captured.\n", cur_inst);
}
```

2. Make simulators for I types

```
void handle_addi(unsigned int cur_inst)
{
    unsigned int rd = MASK11_7(cur_inst), rs1 = MASK19_15(cur_inst);
    int imm12 = sext(MASK31_20(cur_inst), 12);
    NEXT_LATCHES.REGS[rd] = CURRENT_LATCHES.REGS[rs1] + imm12;
}

void handle_slli(unsigned int cur_inst)
{
    /*
     * Lab2-2 assignment
     */
    unsigned int rd = MASK11_7(cur_inst), rs1 = MASK19_15(cur_inst);
    unsigned int imm5 = MASK24_20(cur_inst);
    NEXT_LATCHES.REGS[rd] = CURRENT_LATCHES.REGS[rs1] << imm5;
}

void handle_xori(unsigned int cur_inst)
{
    /*
     * Lab2-2 assignment
     */
    unsigned int rd = MASK11_7(cur_inst), rs1 = MASK19_15(cur_inst);
    int imm12 = sext(MASK31_20(cur_inst), 12);
    NEXT_LATCHES.REGS[rd] = CURRENT_LATCHES.REGS[rs1] ^ imm12;
}

void handle_srlr(unsigned int cur_inst)
{
    /*
     * Lab2-2 assignment
     */
    unsigned int rd = MASK11_7(cur_inst), rs1 = MASK19_15(cur_inst);
    int imm5 = MASK24_20(cur_inst);
    NEXT_LATCHES.REGS[rd] = CURRENT_LATCHES.REGS[rs1] >> imm5;
}

void handle_srai(unsigned int cur_inst)
{
    /*
     * Lab2-2 assignment
     */
    unsigned int rd = MASK11_7(cur_inst), rs1 = MASK19_15(cur_inst);
    int imm5 = MASK24_20(cur_inst);
    int target = CURRENT_LATCHES.REGS[rs1];
    NEXT_LATCHES.REGS[rd] = sext(target >> imm5, 32 - imm5);
}

void handle_ori(unsigned int cur_inst)
{
    /*
     * Lab2-2 assignment
     */
    unsigned int rd = MASK11_7(cur_inst), rs1 = MASK19_15(cur_inst);
    int imm12 = sext(MASK31_20(cur_inst), 12);
    NEXT_LATCHES.REGS[rd] = CURRENT_LATCHES.REGS[rs1] | imm12;
}

void handle_andi(unsigned int cur_inst)
{
    /*
     * Lab2-2 assignment
     */
    unsigned int rd = MASK11_7(cur_inst), rs1 = MASK19_15(cur_inst);
    int imm12 = sext(MASK31_20(cur_inst), 12);
    NEXT_LATCHES.REGS[rd] = CURRENT_LATCHES.REGS[rs1] & imm12;
}

void handle_lui(unsigned int cur_inst)
{
    /*
     * Lab2-2 assignment
     */
    unsigned int rd = MASK11_7(cur_inst);
    int imm12 = (MASK31_12(cur_inst) << 12);
    NEXT_LATCHES.REGS[rd] = sext(imm12, 12) << 12;
}
```

srai need to use sext to fill in the MSB after right shift

3. Two Jumps

```
void handle_jalr(unsigned int cur_inst)
{
    /*
     * Lab2-2 assignment
     */
    unsigned int rd = MASK11_7(cur_inst),
                rs1 = MASK19_15(cur_inst);
    int imm12 = sext(MASK31_20(cur_inst), 12);
    NEXT_LATCHES.REGS[rd] = CURRENT_LATCHES.PC + 4;
    NEXT_LATCHES.PC = sext(imm12, 12) + CURRENT_LATCHES.REGS[rs1];
}

void handle_jal(unsigned int cur_inst)
{
    /*
     * Lab2-2 assignment
     */
    unsigned int rd = MASK11_7(cur_inst);
    int imm12 = (MASK20(cur_inst) << 11) +
                (MASK30_21(cur_inst) << 1) +
                (MASK31(cur_inst) << 20) +
                (MASK19_12(cur_inst) << 12);
    NEXT_LATCHES.REGS[rd] = CURRENT_LATCHES.PC + 4;
    NEXT_LATCHES.PC = sext(imm12, 12) + CURRENT_LATCHES.PC;
}
```

4. Branches

```
void handle_beq(unsigned int cur_inst)
{
    unsigned int rs1 = MASK19_15(cur_inst), rs2 = MASK24_20(cur_inst);
    int imm12 = (MASK31(cur_inst) << 12) +
                (MASK7(cur_inst) << 11) +
                (MASK30_25(cur_inst) << 5) +
                (MASK11_8(cur_inst) << 1);
    if (CURRENT_LATCHES.REGS[rs1] == CURRENT_LATCHES.REGS[rs2])
        NEXT_LATCHES.PC = sext(imm12, 12) + CURRENT_LATCHES.PC;
}

void handle_bne(unsigned int cur_inst)
{
    /*
     * Lab2-2 assignment
     */
    unsigned int rs1 = MASK19_15(cur_inst), rs2 = MASK24_20(cur_inst);
    int imm12 = (MASK31(cur_inst) << 12) +
                (MASK7(cur_inst) << 11) +
                (MASK30_25(cur_inst) << 5) +
                (MASK11_8(cur_inst) << 1);
    if (CURRENT_LATCHES.REGS[rs1] != CURRENT_LATCHES.REGS[rs2])
        NEXT_LATCHES.PC = sext(imm12, 12) + CURRENT_LATCHES.PC;
}

void handle_blt(unsigned int cur_inst)
{
    /*
     * Lab2-2 assignment
     */
    unsigned int rs1 = MASK19_15(cur_inst), rs2 = MASK24_20(cur_inst);
    int imm12 = (MASK31(cur_inst) << 12) +
                (MASK7(cur_inst) << 11) +
                (MASK30_25(cur_inst) << 5) +
                (MASK11_8(cur_inst) << 1);
    if (CURRENT_LATCHES.REGS[rs1] < CURRENT_LATCHES.REGS[rs2])
        NEXT_LATCHES.PC = sext(imm12, 12) + CURRENT_LATCHES.PC;
}

void handle_bge(unsigned int cur_inst)
{
    /*
     * Lab2-2 assignment
     */
    unsigned int rs1 = MASK19_15(cur_inst), rs2 = MASK24_20(cur_inst);
    int imm12 = (MASK31(cur_inst) << 12) +
                (MASK7(cur_inst) << 11) +
                (MASK30_25(cur_inst) << 5) +
                (MASK11_8(cur_inst) << 1);
    if (CURRENT_LATCHES.REGS[rs1] >= CURRENT_LATCHES.REGS[rs2])
        NEXT_LATCHES.PC = sext(imm12, 12) + CURRENT_LATCHES.PC;
}
```

5. Load

```
void handle_lb(unsigned int cur_inst)
{
    unsigned int rd = MASK11_7(cur_inst), rs1 = MASK19_15(cur_inst);
    int imm12 = MASK31_20(cur_inst);
    NEXT_LATCHES.REGS[rd] = sext(MASK7_0(MEMORY[sext(imm12, 12) + CURRENT_LATCHES.REGS[rs1]]), 8);
}

void handle_lh(unsigned int cur_inst)
{
    /*
     * Lab2-2 assignment
     */
    unsigned int rd = MASK11_7(cur_inst), rs1 = MASK19_15(cur_inst);
    int imm12 = MASK31_20(cur_inst);

    unsigned int addr = CURRENT_LATCHES.REGS[rs1] + sext(imm12, 12);

    unsigned int low = MEMORY[addr];
    unsigned int high = MEMORY[addr + 1];
    unsigned int halfword = (high << 8) | low;

    NEXT_LATCHES.REGS[rd] = sext(halfword, 16);
}

void handle_lw(unsigned int cur_inst)
{
    /*
     * Lab2-2 assignment
     */
    unsigned int rd = MASK11_7(cur_inst), rs1 = MASK19_15(cur_inst);
    int imm12 = MASK31_20(cur_inst);
    unsigned int addr = CURRENT_LATCHES.REGS[rs1] + sext(imm12, 12);
    int word = (MEMORY[addr + 3] << 24) | (MEMORY[addr + 2] << 16) | (MEMORY[addr + 1] << 8) | MEMORY[addr];
    NEXT_LATCHES.REGS[rd] = word;
}
```

6. Store

```
void handle_sb(unsigned int cur_inst)
{
    /*
     * Lab2-2 assignment
     */
    unsigned int rs1 = MASK19_15(cur_inst);
    unsigned int rs2 = MASK24_20(cur_inst);
    int imm12 = (MASK31_25(cur_inst) << 5) | MASK11_7(cur_inst);
    unsigned int addr = CURRENT_LATCHES.REGS[rs1] + sext(imm12, 12);
    MEMORY[addr] = NEXT_LATCHES.REGS[rs2] & 0xFF;
}

void handle_sh(unsigned int cur_inst)
{
    /*
     * Lab2-2 assignment
     */
    unsigned int rs1 = MASK19_15(cur_inst);
    unsigned int rs2 = MASK24_20(cur_inst);

    int imm12 = (MASK31_25(cur_inst) << 5) | MASK11_7(cur_inst);

    unsigned int addr = CURRENT_LATCHES.REGS[rs1] + sext(imm12, 12);
    unsigned int value = CURRENT_LATCHES.REGS[rs2] & 0xFFFF;
    MEMORY[addr] = value & 0xFF;
    MEMORY[addr + 1] = (value >> 8) & 0xFF;
}

void handle_sw(unsigned int cur_inst)
{
    /*
     * Lab2-2 assignment
     */
    unsigned int rs1 = MASK19_15(cur_inst);
    unsigned int rs2 = MASK24_20(cur_inst);
    int imm12 = (MASK31_25(cur_inst) << 5) | MASK11_7(cur_inst);

    unsigned int addr = CURRENT_LATCHES.REGS[rs1] + sext(imm12, 12);
    unsigned int value = CURRENT_LATCHES.REGS[rs2];
    MEMORY[addr] = value & 0xFF;
    MEMORY[addr + 1] = (value >> 8) & 0xFF;
    MEMORY[addr + 2] = (value >> 16) & 0xFF;
    MEMORY[addr + 3] = (value >> 24) & 0xFF;
}
```

7. Compile the simulator in terminal

```
CUHK-CENG3420-Lab -- zsh -- 80x24
Last login: Tue Mar 17 11:32:52 on ttys001
kinggor@WongdeMacBook-Air ~ % cd /Users/kinggor/Documents/CENG3420/Lab2/CUHK-CEN
G3420-Lab
kinggor@WongdeMacBook-Air CUHK-CENG3420-Lab % make
make: Nothing to be done for 'all'.
kinggor@WongdeMacBook-Air CUHK-CENG3420-Lab %
```

8.1 Check isa.bin

```
kinggor@WongdeMacBook-Air CUHK-CENG3420-Lab % ./sim ./benchmarks/isa.bin
[INFO]: Welcome to the RISCVC LC Simulator
```

```
[INFO]: read 136 words (544 bytes) from program into memory.
```

```
RISCV LC SIM > go
```

```
[INFO]: simulating...
```

```
[INFO]: cur_inst = 0x00000537
[INFO]: cur_inst = 0x00050513
[INFO]: cur_inst = 0x00052503
[INFO]: cur_inst = 0x00054263
[INFO]: cur_inst = 0x00d50893
[INFO]: cur_inst = 0xff105ee3
[INFO]: cur_inst = 0x00000537
[INFO]: cur_inst = 0x00050513
[INFO]: cur_inst = 0x00150583
[INFO]: cur_inst = 0x7ff5c613
[INFO]: cur_inst = 0x00052503
[INFO]: cur_inst = 0x00a606b3
[INFO]: cur_inst = 0x00a60733
[INFO]: cur_inst = 0xffff8793
[INFO]: cur_inst = 0x00c0006f
[INFO]: cur_inst = 0xff9ff06f
[INFO]: cur_inst = 0x00c000ef
[INFO]: cur_inst = 0x40a88333
[INFO]: cur_inst = 0x00080667
[INFO]: cur_inst = 0x0100006f
[INFO]: cur_inst = 0x00381893
[INFO]: cur_inst = 0x0028d693
[INFO]: cur_inst = 0x40d006b3
[INFO]: cur_inst = 0x4026d713
[INFO]: cur_inst = 0x00f6c693
[INFO]: cur_inst = 0x40d006b3
[INFO]: cur_inst = 0x000004b7
[INFO]: cur_inst = 0x00448493
[INFO]: cur_inst = 0x00d48423
[INFO]: cur_inst = 0x00e6c6b3
[INFO]: cur_inst = 0x00d4a823
[INFO]: cur_inst = 0x000707f
[INFO]: RISCV LC is halted.
```

```
RISCV LC SIM > rd
```

```
current register/bus values:
```

```
-----
instruction count: 32
PC          : 0x00400000
registers:
zero [x0]: 0x00000000
ra [x1]: 0x00000040
sp [x2]: 0x00000000
gp [x3]: 0x00000000
tp [x4]: 0x00000000
t0 [x5]: 0x00000000
t1 [x6]: 0x00000000
t2 [x7]: 0x00000000
fp/s0 [x8]: 0x0000007c
s1 [x9]: 0x00000084
a0 [x10]: 0xffffffff
a1 [x11]: 0xffffffff
a2 [x12]: 0xfffff800
a3 [x13]: 0xffffffee
a4 [x14]: 0xfffffff9
a5 [x15]: 0x0000000a
```

```
RISCV LC SIM > m 0x84 0x84
```

```
memory content [0x00000084..0x00000084]:
```

```
-----
0x00000084 (132) : 0xffffffff7
```

```
RISCV LC SIM > m 0x95 0x95
```

```
memory content [0x00000095..0x00000095]:
```

```
-----
0x00000095 (149) : 0x00ffffff
```

```
RISCV LC SIM > m 0x94 0x94
```

```
memory content [0x00000094..0x00000094]:
```

```
-----
0x00000094 (148) : 0xffffffee
```

```
RISCV LC SIM >
```

8.2 Check count10.bin

```
t2 = 55/0x00000037
```

8.3 Check swap.bin

(memory address: 0x00000034) is 0x1234 and (memory address: 0x00000038) is 0xabcd.

8.4 Check add4.bin

```
RISCV LC SIM > ^C
[kinggor@WongdeMacBook-Air CUHK-CENG3420-Lab % ./sim ./benchmarks/add4.bin
[INFO]: Welcome to the RISCV LC Simulator

[INFO]: read 60 words (240 bytes) from program into memory.

RISCV LC SIM > go

[INFO]: simulating...

[INFO]: cur_inst = 0x000005b7
[INFO]: cur_inst = 0x03458593
[INFO]: cur_inst = 0x0005a583
[INFO]: cur_inst = 0x00000637
[INFO]: cur_inst = 0x03860613
[INFO]: cur_inst = 0x00062603
[INFO]: cur_inst = 0xfff58593
[INFO]: cur_inst = 0x00160613
[INFO]: cur_inst = 0xfe059ce3
[INFO]: cur_inst = 0xfff58593
[INFO]: cur_inst = 0x00160613
[INFO]: cur_inst = 0xfe059ce3
[INFO]: cur_inst = 0xfff58593
[INFO]: cur_inst = 0x00160613
[INFO]: cur_inst = 0xfe059ce3
[INFO]: cur_inst = 0xfff58593
[INFO]: cur_inst = 0x00160613
[INFO]: cur_inst = 0xfe059ce3
[INFO]: cur_inst = 0x000006b7
[INFO]: cur_inst = 0x03860693
[INFO]: cur_inst = 0x00c6a023
[INFO]: cur_inst = 0x0000707f
[INFO]: RISCV LC is halted.

RISCV LC SIM > m 0x38 0x38

memory content [0x00000038..0x00000038]:
-----
0x00000038 (56) : 0xffffffff
-----
-
```

(memory address: 0x00000038) is -1 (0xffffffff)